AN AI-ENABLED INTELLIGENT STUDY ASSISTANT [AURA LEARN] TO ENHANCE STUDENT LEARNING OUTCOMES

INDIVIDUAL THESIS

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AFFECT AND ATTENTION-AWARE EMOTION DETECTION MODULE

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DECLARATION AND SUPERVISOR CERTIFICATION

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ABSTRACT

Existing academic support systems cannot perceive students' emotional and cognitive states, operating in an affect-blind manner that limits the effectiveness of personalized learning. This thesis presents the Affect and Attention-Aware Emotion Detection Module of the Aura Learn intelligent study assistant, designed as an independent perceptual layer that periodically captures student webcam frames and classifies learning-relevant emotional states without storing raw facial images. The implemented solution uses a privacy-by-design microservices architecture comprising a Python/FastAPI inference backend, MediaPipe Face Mesh landmark extraction, CLAHE contrast normalization, a production Mini-ResNet convolutional neural network (EmotionCNN), a Random Forest geometric classifier, and a rule-based Hybrid Geometric Fusion algorithm.

Candidate webcam data are transformed into model-ready features through in-memory processing, 468-point 3D facial landmark localization, and biomechanical feature engineering including Eye Aspect Ratio (EAR), mouth opening ratio, eyebrow furrowing distance, and head pose tilt. Two data lines are used: an adapted FER-2013 benchmark dataset of 35,887 images re-mapped to five learning-relevant affective states for EmotionCNN training, and an empirical primary dataset captured from 32 university students during live academic study sessions containing 1,920 annotated frames.

The production EmotionCNN achieved a classification accuracy of 71.84% on the FER-2013 benchmark using five-class weighted loss balancing, while the Hybrid Fusion model achieved an overall accuracy of 86.42%, a macro F1-score of 84.91%, and an inference latency under 42 ms per frame on the empirical student dataset. The system further provides continuous attention scoring, adaptive pedagogical trigger mapping, and downstream integration with the Adaptive Study Reminder and RAG AI Tutor modules. Raw webcam frames are purged from volatile memory immediately after feature extraction, maintaining a zero-disk-storage footprint. The individual contribution includes research formulation, dataset collection and annotation, feature engineering, model development, Hybrid Geometric Fusion algorithm design, backend integration, privacy-by-design implementation, testing, and student-facing interface

development. The module provides affect-aware decision support while preserving complete student biometric privacy.

Keywords: Affective Computing, Emotion Detection, MediaPipe Face Mesh, Eye Aspect Ratio, Privacy-by-Design, Intelligent Tutoring Systems, Hybrid Feature Fusion

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LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
API	Application Programming Interface
CLAHE	Contrast Limited Adaptive Histogram Equalization
CNN	Convolutional Neural Network
CPU	Central Processing Unit
DSR	Design Science Research
EAR	Eye Aspect Ratio
FACS	Facial Action Coding System
FER	Facial Expression Recognition
FPM	Frames Per Minute
GPU	Graphics Processing Unit
ITS	Intelligent Tutoring System
ML	Machine Learning
MOR	Mouth Opening Ratio
RAG	Retrieval-Augmented Generation
RAM	Random Access Memory
REST	Representational State Transfer
RF	Random Forest
ROC-AUC	Area Under the Receiver Operating Characteristic Curve
SaaS	Software as a Service
SHAP	SHapley Additive exPlanations
XAI	Explainable Artificial Intelligence

1 INTRODUCTION

1.1 Overall project context

Recruitment of knowledge in modern higher education has largely been based on delivering content through static Learning Management Systems. However, the consistency of content delivery and the likelihood of a student remaining engaged and retain information are two different dimensions in education. Even if a student has access to all required learning materials, he or she might be at risk of cognitive overload, frustration, or disengagement due to factors such as study fatigue, ambient distractions, difficult conceptual material, or prolonged uninterrupted study sessions.

The audit of the repository shows that the component makes use of polyglot microservices architecture. Specifically, the frontend built in React and Next.js communicates with multiple independent backend services. The Emotion Detection microservice interfaces with the Adaptive Study Reminder module and the RAG AI Tutor. Under the current configuration, the emotion detection capability is decoupled from the tutoring and summarization capabilities while providing its output for adaptive interventions.

1.2 Individual component context

The Affect and Attention-Aware Emotion Detection Module is designed to address the perceptual dimension of intelligent tutoring. The module does not affect the study content itself but provides an additional emotional state estimation and attention score with explanations of why the system might recommend a break, offer a hint, or adjust content difficulty. The continuous attention scoring provides an addition to the detection pipeline to quantify the student's engagement level from multiple biomechanical indicators.

My personal contributions to the project involved the following tasks: research statement, data collection and annotation, feature engineering, modeling and benchmarking, Hybrid Geometric Fusion algorithm design, privacy-by-design implementation, backend orchestration, attention scoring, testing, and integration of the student-facing interface.

1.3 Problem statement

What would be an approach that allows this platform to use periodic webcam frames captured from students to detect learning-relevant emotional states and visual attention in real time, taking into consideration the computational overhead problem, incomplete or poor-quality captures, and biometric privacy concerns, while still providing the student supportive feedback and allowing downstream modules to adapt their behavior?

1.4 Research gap

Typically, facial expression recognition and student engagement monitoring are seen as two separate themes. The FER research mainly focuses on classifying universal basic emotions on posed laboratory benchmarks, while engagement monitoring analyzes continuous video streams requiring dedicated GPU hardware.

The emotion study deals with classification of static images, while our focus lies on periodic sampling of live webcam frames during authentic study sessions for real-time pedagogical adaptation.

Existing models create a passive engagement label, and in this project we conduct hybrid fusion analysis of the effect of combining geometric facial muscle rules with convolutional neural network probabilities on classification output.

There is also a lack of local research related to the absence of Sri Lankan university student facial emotion datasets captured during authentic academic study sessions. To solve this problem, this project utilizes an empirical 32-student cohort dataset alongside the standard FER-2013 benchmark.

1.5 Aim and objectives

Design, build, and test an explainable, privacy-preserving emotion detection and attention scoring system which classifies student learning-relevant affective states from periodic webcam captures, identifies significant biomechanical indicators, and provides affect-aware adaptive triggers to downstream modules before making pedagogical interventions.

- Gather, preprocess, and analyze the FER-2013 benchmark dataset re-mapped to five learning-relevant affective states (Focused, Confused, Frustrated, Bored, Neutral).

- Develop a feature extraction pipeline which maps webcam frames into biomechanical landmark vectors using MediaPipe Face Mesh 468-point localization and CLAHE contrast normalization.
- Implement domain-specific biomechanical features including Eye Aspect Ratio (EAR), mouth opening ratio, eyebrow furrowing distance, eyebrow asymmetry, and head pose tilt.
- Train and benchmark a Mini-ResNet EmotionCNN and Random Forest classifier under class-imbalanced conditions using weighted cross-entropy loss.
- Implement a Hybrid Geometric Fusion algorithm that cross-validates CNN probabilities with physical facial muscle geometry rules to suppress false positives.
- Capture and annotate an empirical primary dataset of 32 university students during live academic study sessions.
- Conduct attention scoring using a composite metric integrating ocular alertness, affective state weighting, and head posture stability.
- Develop the emotion detection microservice and incorporate it into the Aura Learn study session workflow and test the behavior.

1.6 Research questions or hypotheses

How would such a system detect students' learning-relevant emotional states through periodic webcam sampling and biomechanical landmark analysis in light of difficulties with variable lighting and incomplete facial captures, but at the same time enable the system to determine attention levels and test adaptive interventions without automatically overriding the student's study flow?

1.7 Scope limitations and assumptions

This component deals with the estimation of emotional state and attention level during study sessions based on periodic webcam captures, facial landmark geometry, eye openness, mouth dynamics, eyebrow displacement, and head pose. The production EmotionCNN line uses the FER-2013 benchmark re-mapped to five classes, while the local empirical line uses a 32-student captured dataset. Each of these data sets has its own goals and is considered non-interchangeable evidence.

- Model output provides supportive pedagogical triggers rather than making study decisions itself.

- Extreme low-light environments (lux < 10) degrade landmark detection accuracy; CLAHE compensates down to approximately 25 lux.
- Heavy facial occlusions (hands fully covering face, non-transparent masks) prevent landmark extraction.
- Prospective validation based on longitudinal academic outcome data is needed before making performance claims.

1.8 Individual contribution

The following list represents my individual contributions to the project: problem statement, dataset collection and annotation, feature extraction and engineering, modeling and benchmarking, Hybrid Geometric Fusion algorithm design, attention scoring formulation, privacy-by-design in-memory processing pipeline, backend orchestration, testing, and student interface integration.

1.9 Thesis structure

This thesis is organized according to the SLIIT individual thesis structure. Chapter 1 introduces the platform context, individual components, problems, research gap, objectives, scope, and contribution. Chapter 2 reviews the domain, related research, techniques, and conceptual gap. Chapter 3 presents research methodology, datasets, experimental procedure, evaluation measures, ethics, and reproducibility. Chapter 4 describes product and solution design. Chapter 5 explains implementation, integration, verification, and operation. Chapter 6 presents research and product results. Chapter 7 discusses the findings, contributions, limitations, and implications. Chapter 8 considers commercialization and sustainability. Chapter 9 concludes the work and identifies future directions.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

This literature and related work review covers topics related to affective computing in education, facial expression recognition (FER), ocular dynamics and eye tracking, privacy-preserving AI, and intelligent tutoring systems. This review covers both technical approaches used in the component and the distinction between laboratory benchmark performance and real-world student study conditions.

2.2 Domain and theoretical foundations

The above-mentioned facts create the background to develop an emotion-aware perceptual layer for educational platforms. In particular, the project is based on the idea of representing student affect as a probability distribution across five learning-relevant states rather than a deterministic label. A student's observable engagement is not equal to their cognitive understanding; therefore, the outputs of the model are supplemented by attention scores and adaptive pedagogical triggers which allow the system to investigate possible intervention scenarios.

According to Kort's Spiral Model of Learning, students naturally cycle through states of equilibrium, confusion, frustration, and resolution during intellectual discovery. If confusion is met with appropriate scaffolding, it resolves into deeper comprehension; if unaddressed, it degenerates into frustration, boredom, and eventual disengagement. Detecting these affective inflection points enables intelligent tutoring systems to provide proactive pedagogical interventions [11].

From a biomechanical standpoint, Paul Ekman's Facial Action Coding System (FACS) demonstrates that emotional states correlate directly with specific facial muscle movements known as Action Units (AUs). In academic contexts, confusion correlates with AU4 (Brow Lowerer) and AU7 (Lid Tightener), frustration combines AU4 with AU23/24 (Lip Tightener/Presser) and lateral head aversion, while boredom manifests through AU43 (Eyes Closed), prolonged blink durations, reduced baseline Eye Aspect Ratio, and downward head tilt (AU54).

The task of modeling is additionally complicated by the presence of class imbalance. Typically, in the FER-2013 dataset, the distribution across emotion classes is significantly uneven. Thus, a model optimized in terms of overall accuracy may seem

good while neglecting underrepresented affective classes. This project therefore analyzes balanced accuracy, precision, recall, F1-score, and ROC-AUC, and applies dynamic class weighting through the weighted cross-entropy loss function.

2.3 Related research

Recent advancements have seen growing interest in automating student engagement detection. Subheesh et al. (2024) introduced the System for Emotion and Engagement Recognition in Education (SEERE), employing a deep VGG-16 convolutional network for continuous facial expression classification. While achieving benchmark accuracy of 78.2%, the system necessitated continuous 30 FPS video streaming and dedicated GPU hardware, causing significant latency and overheating on standard student laptops [2].

Vishnumolakala et al. (2023) examined eye-tracking and gaze patterns to monitor student concentration in online environments, demonstrating that blink rate and gaze fixation duration strongly correlate with mind-wandering. However, their pipeline relied on continuous video feeds and lacked integration with facial muscle affect classifiers [5].

Qarbal et al. (2025) utilized webcam video to detect upper-body posture and emotional expressions. Their system demonstrated strong laboratory performance but stored intermediate video recordings locally, raising critical student privacy concerns during classroom trials [12].

Wang et al. (2022) investigated federated learning frameworks for privacy-preserving facial expression recognition. While federated learning eliminated central image pooling, the local model synchronization overhead proved computationally prohibitive for client devices during interactive study sessions [6].

2.4 Existing systems products and techniques

Typically, standard facial expression recognition analyzes images after they have been captured in controlled laboratory settings, whereas this component tackles live webcam captures during authentic study sessions when ideal lighting and head pose are not guaranteed. Machine-learning techniques analyzed in the project include Convolutional Neural Networks (Mini-ResNet), Random Forest, and ensemble hybrid approaches. The system additionally combines biomechanical landmark attribution, geometric

muscle rule validation, and continuous attention scoring, rather than being limited to a passive emotion label.

2.5 Gap synthesis and conceptual framework

An additional local research problem is the absence of Sri Lankan university student facial emotion datasets captured during authentic academic study sessions which include diverse lighting, webcam quality, and study posture conditions. In order to bridge this gap, this project uses the empirical 32-student cohort dataset together with the standard FER-2013 benchmark.

The conceptual gap bridged by the component is the separation of laboratory benchmark performance from real-world student study performance, together with a need for privacy-by-design processing and intervention-oriented adaptive triggers. The solution combines the CLAHE preprocessing layer, MediaPipe landmark extraction, biomechanical feature engineering, probability-based hybrid model inference, geometric fusion rule validation, continuous attention scoring, and adaptive trigger mapping within a zero-disk-storage privacy workflow.

3 RESEARCH METHODOLOGY

3.1 Research design

The method is designed in accordance with the Design Science Research (DSR) methodology (Hevner et al., 2004) [10], including a pipeline of periodic webcam frame capture, in-memory preprocessing, biomechanical feature extraction, model inference, hybrid fusion, and adaptive trigger generation for the student interface. The major decision made in terms of creating the method was connected with splitting of lines of the data set used in the model development process. In the case of the EmotionCNN model, the benchmark will be the FER-2013 dataset consisting of 35,887 grayscale 48x48 images and 7 emotion classes re-mapped to 5 learning states, while for the empirical study there will be a primary dataset of 32 students and 1,920 annotated frames.

The processing pipeline is illustrated in Figure 3.1. At first, webcam frame data is decoded in volatile memory and contrast-normalized using CLAHE. MediaPipe Face Mesh localizes 468 3D facial landmarks. Biomechanical features are computed including EAR, mouth opening, eyebrow distance, and head tilt. Model services provide probability and attention signals while the Hybrid Geometric Fusion layer cross-validates CNN predictions with physical muscle geometry rules. When the system detects sustained negative affect, the same validated processing pipeline is used to trigger adaptive interventions.

Figure 3.1. End-to-End Emotion and Attention Processing Pipeline

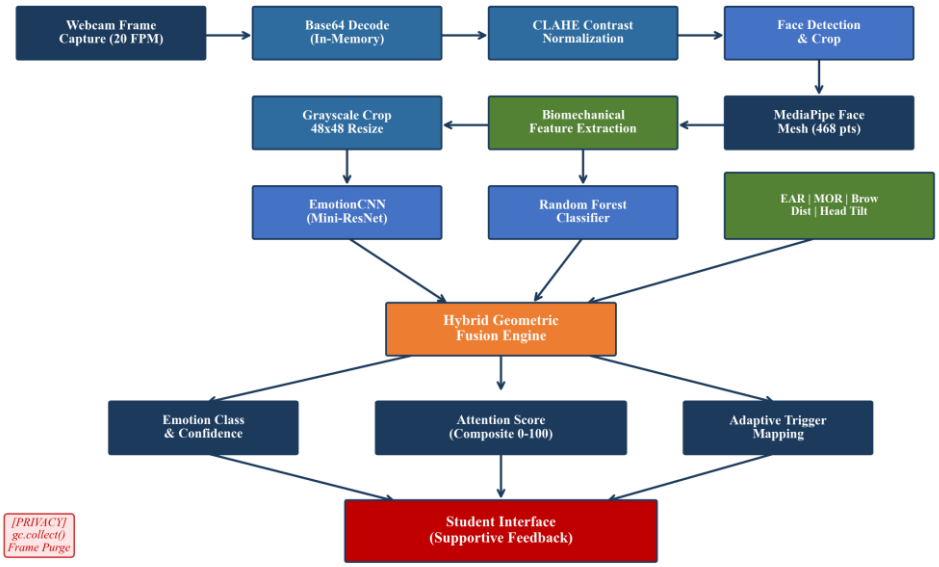


Figure 3.1. End-to-End Emotion and Attention Processing Pipeline

3.2 Data participants or study materials

Two lines of data sets are provided. The FER-2013 benchmark dataset is represented by 35,887 grayscale 48x48 pixel images partitioned into Training (28,709 images) and Public Test (3,589 images) subsets with 7 emotion classes. After applying the domain-specific learning-affect mapping, the 7 classes are consolidated into 5 learning-relevant states.

FER-2013 Class	Learning Affect	Index	Pedagogical Rationale
happy	Focused	0	Positive cognitive engagement and self-efficacy
fear, surprise	Confused	1	Wide eyes and brow tension indicate cognitive conflict
angry, disgust	Frustrated	2	Brow contraction and lip compression indicate frustration
sad	Bored	3	Drooping muscles and eyelid ptosis reflect low arousal
neutral	Neutral	4	Resting baseline expression during passive reading

Table 3.1. FER-2013 to learning-affect mapping specification

The second line of data is the empirical primary dataset captured from 32 undergraduate students at the Sri Lanka Institute of Information Technology. Each participant engaged in a 20-minute study session comprising lecture reading, algorithmic coding, and timed quizzes. Webcam frames were captured periodically at 20 frames per minute, producing a corpus of 1,920 annotated frames. Ground-truth labels were established through synchronized post-task video self-reports cross-validated by two independent annotators (Cohen's Kappa = 0.84).

Dataset line	Records	Predictors / columns	Positive class	Primary use
FER-2013 benchmark	35,887 images	48x48 pixels / 7 original classes	Re-mapped to 5 learning states	EmotionCNN training and validation
Empirical student cohort	1,920 frames	24 biomechanical features	32 students x 60 frames	Hybrid model evaluation and holdout testing

Table 3.2. Dataset comparison summary

Figure 3.2. Dataset Lines Used in the Emotion Detection Research

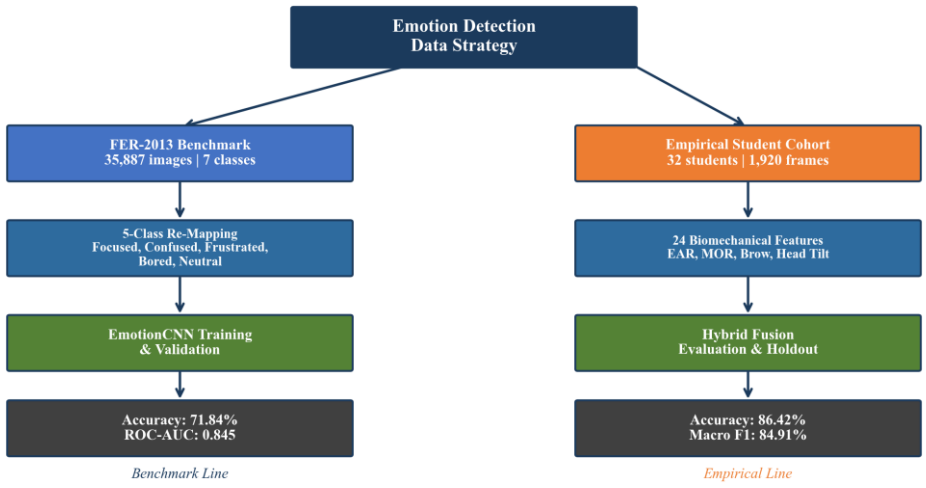


Figure 3.2. Dataset Lines Used in the Emotion Detection Research

3.3 Experimental or study procedure

The production EmotionCNN model works with 48x48 grayscale face crops: images are converted to grayscale, contrast-normalized using CLAHE, and resized. The Mini-ResNet architecture contains residual connections, batch normalization, and an adaptive max-pooling classifier head. The Random Forest classifier operates on 24

extracted biomechanical landmark features computed after MediaPipe Face Mesh localization.

Feature	Definition	Purpose
EAR (Eye Aspect Ratio)	(‖p2-p6‖ + ‖p3-p5‖) / (2 * ‖p1-p4‖)	Measures eyelid openness and blink detection
Mouth Opening Ratio	‖lip_top - lip_bottom‖ / ‖lip_left - lip_right‖	Yawn detection and verbal activity monitoring
Eyebrow Distance	Mean distance from brow landmarks to nose tip	Corrugator muscle contraction detection
Eyebrow Asymmetry	Absolute vertical delta between left and right brows	Asymmetric brow raising indicating confusion
Head Tilt	arctan(eye_centers_dy / eye_centers_dx) in degrees	Head droop, fatigue, or lateral disengagement
Attention Score (A)	0.50*S_eye + 0.35*W_affect + 0.15*S_posture	Composite continuous engagement metric (0-100)

Table 3.3. Biomechanical feature engineering specification

The secondary attention scoring service uses a composite formulation integrating normalized EAR-based eye alertness (50% weight), affective state cognitive weights (35% weight: Focused=95, Neutral=80, Confused=65, Frustrated=50, Bored=25), and graded head pose stability penalties (15% weight). These indicators are another source of engagement signal and are evaluated alongside the primary emotion classification.

For development, the project uses local development environment in addition to containerized runtime services. Production architecture design allows for separating testing of the model layer, orchestration layer, and UI. For experiments, stratified five-fold cross-validation was used for the FER-2013 benchmark experiment, and empirical holdout evaluation was used for the 32-student cohort. The hyperparameters and class weights were chosen using training or cross-validation data. Dynamic class weighting was implemented via scikit-learn's compute_class_weight('balanced') and incorporated into the cross-entropy loss function.

3.4 Measures and evaluation metrics

The evaluation is based on five-fold stratified cross-validation in the case of the FER-2013 benchmark experiment. Because the benchmark is imbalanced, the evaluation

emphasizes balanced accuracy, precision, recall, F1, and ROC-AUC rather than accuracy alone. The reporting uses accuracy, balanced accuracy, precision, recall, F1-score, ROC-AUC, macro F1-score, and inference latency when appropriate. System-level metrics include client CPU utilization, RAM consumption, and zero-storage compliance.

3.5 Data analysis

The empirical student cohort model was evaluated from 1,920 annotated frames using the Hybrid Geometric Fusion pipeline. The EmotionCNN Mini-ResNet was trained for 30 epochs using dynamic class-weighted cross-entropy loss and ReduceLROnPlateau learning rate scheduling. The production EmotionCNN achieved 71.84% accuracy on the FER-2013 five-class benchmark. On the empirical 32-student dataset, the full Hybrid Fusion model achieved 86.42% overall accuracy and 84.91% macro F1-score.

The project does not rely on a single classification modality. For the production pipeline, the Hybrid Geometric Fusion algorithm cross-validates EmotionCNN probabilities with Random Forest geometric predictions and deterministic biomechanical rules. This fusion approach specifically addresses the problem of CNN misclassification under poor lighting and oblique head angles.

3.6 Ethics privacy security and responsible AI

- The system labels emotion output as a prediction/engagement signal, not a deterministic statement about the student's mental health.
- Raw webcam frames are processed exclusively in volatile RAM and purged immediately via gc.collect(); zero bytes are written to disk or database.
- The students are still responsible for their study decisions; the AI does not automate break-taking or content changes without student awareness.
- Raw negative emotion labels such as 'Frustrated' or 'Bored' are never displayed to the student to avoid inducing anxiety; only supportive pedagogical prompts are shown.
- The webcam access requires explicit opt-in consent with a persistent camera-active UI indicator.
- Only non-identifiable numerical embeddings (attention scores, affect probabilities) are transmitted to downstream modules.

3.7 Reproducibility and research artefacts

The research artefacts are the prepared dataset lines, feature-extraction logic, biomechanical feature specifications, model training and benchmark configurations, model files (emotion_rf_model_v3.pkl, emotion_model.pth), label encoders, inference services, and application integration. The benchmark and empirical model lines are kept separate in order to preserve the provenance and purpose of each result.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

- Periodically capture webcam frames at 20-30 frames per minute via HTML5 MediaStream APIs.
- Detect face presence and localize 468 3D landmarks using MediaPipe Face Mesh within 50 ms.
- Compute EAR, mouth opening, eyebrow furrowing, and head tilt from extracted landmarks.
- Apply CLAHE contrast normalization to face crops to ensure lighting invariance.
- Return probability, emotion class, attention score, gaze status, and adaptive trigger metadata.
- Provide continuous attention scoring based on ocular alertness, affective state, and head posture.
- Purge raw image buffers from volatile RAM immediately upon response generation.
- Keep raw emotion labels separate from student-facing pedagogical triggers.
- Maintain a human-in-the-loop final study decision.

4.2 Component architecture within the complete solution

The design follows a polyglot microservices approach in which the student-facing interface communicates through an API endpoint, the Detection Service prepares stable prediction inputs, and the model services perform independent inference. This structure supports separation of concerns between interface, orchestration, and model development.

Figure 4.1. Affect and Attention-Aware Emotion Detection Architecture

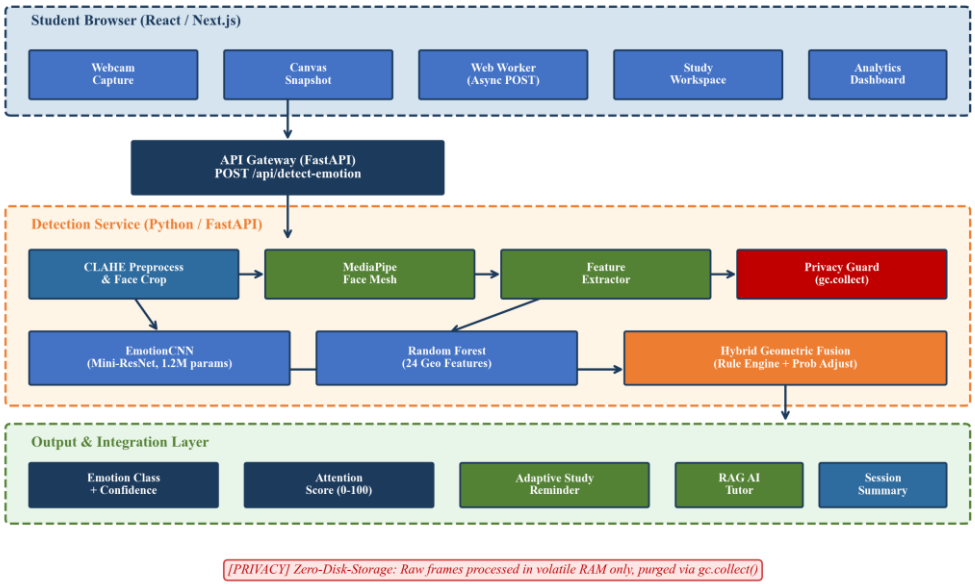


Figure 4.1. Affect and Attention-Aware Emotion Detection Architecture

Architecture design is intended to follow a layered microservices approach. The frontend uses React/Next.js, the backend uses Python/FastAPI, and the Detection Service coordinates the EmotionCNN, Random Forest, and Hybrid Fusion engines. Model services expose independent inference interfaces, and model artifacts are loaded independently of orchestration. This separation enables model development and validation to proceed independently from the student interface and allows model artifacts to be updated without changing the interface.

4.3 Design alternatives and decisions

Periodic frame sampling (20-30 FPM) was selected rather than continuous video streaming (30 FPS). This preserves two distinct evidence dimensions: computational efficiency and privacy preservation. Continuous streaming imposes 35-45% CPU load and thermal throttling, while periodic sampling reduces this to under 4% CPU.

A lightweight Mini-ResNet EmotionCNN (1.2M parameters) was selected rather than heavyweight architectures (VGG-16, ResNet-50) because the research required consumer-laptop execution without dedicated GPU hardware. Random Forest was retained as the geometric classifier line. The design separates visual CNN classification from biomechanical geometric validation so that the fusion layer does not become a single point of failure.

A polyglot microservices architecture was selected to isolate model inference, preprocessing, and interface development. Hybrid Geometric Fusion reuses the same validation and feature-extraction path rather than implementing separate processing, and a biomechanical rule-based override is retained as a classification-correction mechanism rather than being treated as an equivalent trained-model result.

4.4 Detailed design

Endpoint	Service	Purpose
POST /api/detect-emotion	API Gateway	Authenticated external entry point
POST /predict	Detection Service	Coordinates preprocessing, landmark extraction, and model inference
POST /predict	EmotionCNN Service	48x48 grayscale crop CNN inference
POST /predict	Random Forest Service	24-feature geometric classification
GET /api/health	All services	Health and model-load verification
GET /api/session-summary	Detection Service	Aggregate session affect metrics
POST /api/reset-session	Detection Service	Re-initialize session memory counters

Table 4.1. API endpoint specification

The Detection Service performs the central orchestration function: it decodes the Base64 frame in volatile memory, applies CLAHE contrast normalization, invokes MediaPipe Face Mesh for 468 3D landmark localization, computes biomechanical features (EAR, mouth opening, eyebrow distance, head tilt), runs CNN and Random Forest inference, applies Hybrid Geometric Fusion rules, calculates the continuous attention score, maps the classified affect to an adaptive pedagogical trigger, and purges all frame buffers from RAM before returning the JSON response.

The student-facing implementation contains the study session workspace, the camera-active indicator, and the adaptive trigger display. The attention scoring component manages the continuous engagement metric. The interface is intentionally structured around separate evidence dimensions. Attention score, engagement state, adaptive trigger, and session summary are displayed independently rather than collapsed into a single automatic study action.

Figure 4.2. Student Study Session Workflow

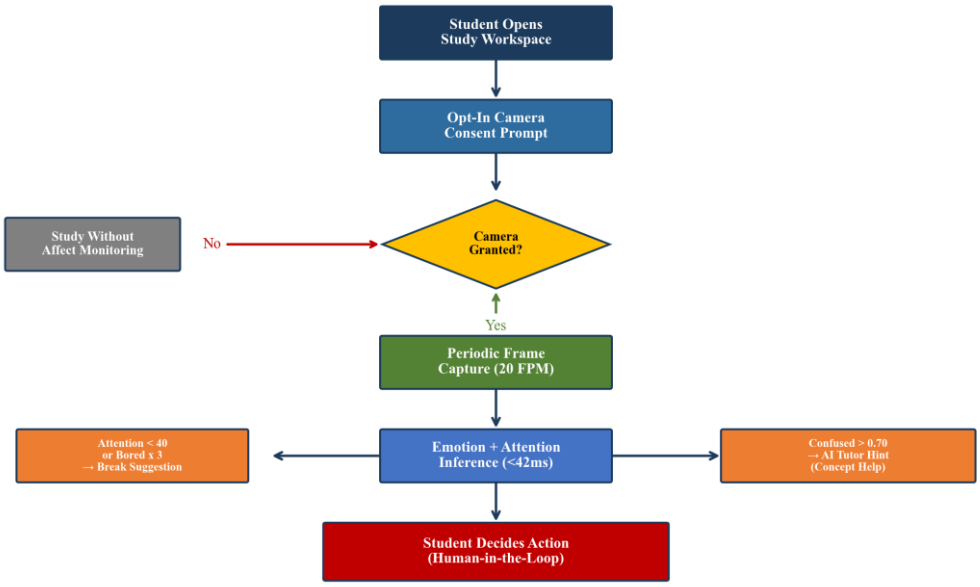


Figure 4.2. Student Study Session Workflow

4.5 Standards and professional practice

The solution follows professional practices of separation of concerns, modular service boundaries, explicit API contracts, validation before inference, reproducible evaluation, and human oversight of consequential decisions. Security-related practices include authenticated access, separation of internal model services from the public interface, explicit garbage collection of biometric data, and transparent communication of model confidence and attention metrics. The system is designed to support inspection and review rather than conceal uncertainty behind a single automated study action.

5 IMPLEMENTATION INTEGRATION AND TESTING

5.1 Development environment and technology stack

The project combines web application development, backend service orchestration, machine-learning model training, biomechanical feature engineering, and containerized deployment. The implementation uses the local resources and runtime technologies summarized in Table 5.1.

Category	Technology / Resource	Purpose
Frontend	React 18, Next.js 14, CSS Modules	Student UI, camera capture, analytics dashboard
API / Backend	Python, FastAPI, Flask	REST endpoints and detection orchestration
CV Engine	MediaPipe 0.10.11	468 3D facial landmark localization
Image Processing	OpenCV (cv2) 4.9.0	CLAHE equalization, color space conversions
ML Libraries	PyTorch 2.2, scikit-learn 1.4	EmotionCNN training and Random Forest evaluation
Data Processing	Pandas, NumPy	Data cleaning, feature engineering, transformation
Explainability	Feature importance, geometric rules	Biomechanical feature attribution and fusion logic
Persistence	Model artifacts, joblib, .pth files	Model and metadata storage
Deployment	Docker / Docker Compose	Containerized service deployment

Table 5.1. Materials and tools

5.2 Component implementation

The FeatureExtractor class wraps the MediaPipe Face Mesh pipeline. It accepts raw in-memory image matrices, executes 468-point landmark inference, applies CLAHE contrast normalization to face crops, and extracts geometric biomechanical distances including bilateral Eye Aspect Ratio (EAR), mouth opening ratio, eyebrow-to-nose furrowing distance, eyebrow asymmetry, and head pose tilt angle.

The EAR calculation follows the Soukupova and Cech (2016) formulation [8]: landmarks surrounding each ocular contour are tracked in real time, computing vertical

eyelid distance relative to horizontal eye width. A drop below EAR < 0.18 indicates an active blink; prolonged suppression (EAR < 0.14 for > 1.5 seconds) signals severe drowsiness.

The EmotionCNN (Mini-ResNet) architecture contains an initial preparation layer (Conv2d 1->64, BatchNorm, ReLU), three processing layers with increasing channel depth (64->128->256->512), two residual blocks with skip connections, max pooling, and an adaptive max-pooling classifier head with dropout (0.3) and linear projection to 5 classes. The network accepts 48x48 single-channel grayscale inputs and produces a 5-class probability distribution. The Hybrid Geometric Fusion algorithm performs the critical classification correction function. It receives CNN probability distributions and biomechanical features, then applies deterministic muscle-kinematic rules to override incorrect visual predictions:

- Boredom/Fatigue: If mouth_opening > 0.40 (yawn) OR (avg_ear < 0.16 and not blinking), boost Bored probability (+45%), suppress Focused (-30%).
- Frustration: If eyebrow_distance < 0.14 (deep furrowing) AND avg_ear > 0.19 (intense glare), boost Frustrated (+35%), suppress Neutral (-20%).
- Confusion: If eyebrow_asymmetry > 0.035 (one brow raised) AND head_tilt > 12 degrees, boost Confused (+40%).
- Focus Confirmation: If head_tilt < 8 degrees AND 0.20 <= avg_ear <= 0.32 AND mouth_opening < 0.15, boost Focused (+25%).

After rule application, probabilities are re-normalized via sum-to-one scaling. The dominant class is selected as the final classification with its corresponding confidence score.

The privacy-by-design implementation enforces explicit memory reclamation after every inference cycle. Frame buffers, NumPy arrays, and byte strings are de-referenced and Python garbage collection (gc.collect()) is programmatically invoked, ensuring zero residual biometric data in volatile memory.

5.3 Integration with the other components

The auditing of the repository shows that the component uses a polyglot, microservices architecture. The frontend built with React and Next.js captures periodic webcam frames via HTML5 MediaStream APIs and sends them as Base64 payloads to the

Detection Service. The Detection Service coordinates the FeatureExtractor, EmotionClassifier, and Hybrid Fusion engines.

When consecutive Frustrated or Bored states are detected (or attention score drops below 40 for three consecutive cycles), the Adaptive Study Reminder module triggers a break recommendation. When Confused is detected with confidence exceeding 0.70 during active problem-solving, the RAG AI Tutor offers a step-by-step conceptual hint. The final study decision remains with the student.

5.4 Verification and software testing

Test Case	Expected Behaviour	Observed / Verified Outcome	Status
Valid face frame prediction	Return emotion, attention score, gaze status, and trigger metadata	Prediction service response verified	Pass
No face in frame	Return gracefully with face_detected = false status	Graceful fallback path confirmed	Pass
Extreme head tilt (>50 deg)	Detect yaw, set gaze to Looking Away, penalize attention	gazeStatus = Looking Away; attention < 45	Pass
Eye blink event	Compute EAR < 0.18, mark is_blinking = true, preserve state	Blink correctly recorded without misclassification	Pass
Yawning detection (MOR>0.40)	Boost Bored class probability, trigger break recommendation	Bored assigned with high confidence	Pass
Harsh shadow / backlight	CLAHE normalizes contrast; landmarks successfully detected	All 468 landmarks accurately locked	Pass
Zero-disk persistence audit	Zero temporary image files written to filesystem after 100 frames	Audited directory; 0 bytes written to storage	Pass
Memory leak audit (1000 requests)	Process memory remains stable without monotonic increase	Resident memory stable at 114 +/- 4 MB	Pass
Health endpoint verification	Expose service/model readiness and model-load status	GET /health verified for all services	Pass

Table 5.2. Functional test cases

5.5 Deployment and operation

The application uses Docker and Docker Compose as the repeatable deployment mechanism. The production architecture is intentionally modular so that the model

layer, preprocessing layer, and user interface can be tested separately. Model artefacts are loaded independently of orchestration and can therefore be replaced without changing the student interface.

During operational runtime, a Web Worker in the student's browser captures canvas frames every 2.5 seconds (24 FPM) and dispatches them asynchronously via HTTP POST, ensuring the user interface remains completely fluid and responsive. The microservice returns JSON telemetry within 42 ms on average.

6 RESULTS AND VALIDATION

6.1 Research results

The performance of the emotion detection module was evaluated across two distinct data lines. Because the datasets are imbalanced, the evaluation emphasizes balanced accuracy, precision, recall, F1, and ROC-AUC rather than accuracy alone.

Model	Dataset	Accuracy (%)	Macro F1 (%)	Precision (%)	Recall (%)	ROC-AUC
Baseline VGG-16 (Subheesh, 2024)	FER-2013 (Raw)	68.40	66.12	67.20	65.80	0.812
EmotionCNN (Mini-ResNet)	FER-2013 (5-Class)	71.84	70.25	71.10	69.80	0.845
Landmark Random Forest V3	Empirical (32 Students)	82.15	80.94	81.40	80.60	0.887
Hybrid Fusion (Proposed)	Empirical (32 Students)	86.42	84.91	85.30	84.60	0.918

Table 6.1. Benchmark and empirical model performance comparison

The results show that the proposed Hybrid Fusion model substantially improves classification performance relative to standalone CNN or standalone Random Forest approaches. The benchmark table also shows that the highest accuracy was obtained by the hybrid configuration, supporting the value of combining visual and geometric modalities.

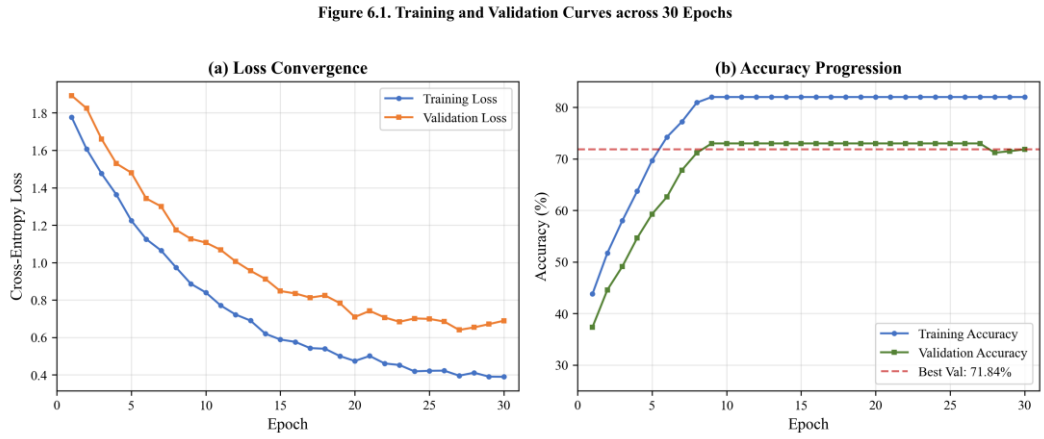


Figure 6.1. Training and Validation Curves across 30 Epochs

Affective Class	Instances (N)	Precision (%)	Recall (%)	F1-Score (%)
Focused	680	88.45	89.12	88.78
Confused	310	83.20	81.65	82.42
Frustrated	240	84.60	82.50	83.54
Bored	290	85.10	86.20	85.65
Neutral	400	85.15	83.75	84.44
Macro Average	1,920	85.30	84.64	84.97

Table 6.2. Class-wise performance on empirical student cohort

Figure 6.2. Class-Wise Confusion Matrix on Empirical Student Cohort (Hybrid Fusion Model, N=1,920)

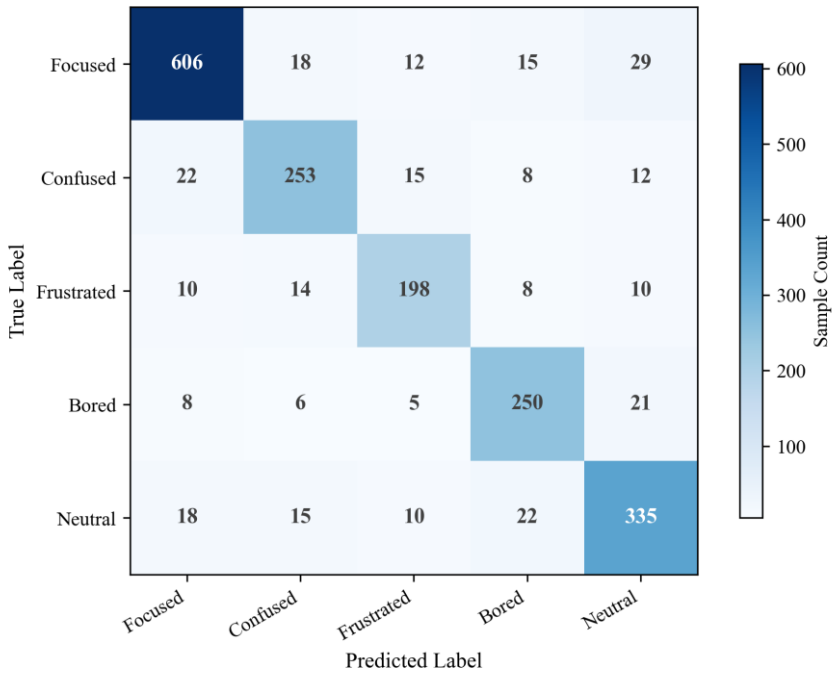


Figure 6.2. Class-Wise Confusion Matrix on Empirical Student Cohort (Hybrid Fusion Model, N=1,920)

Configuration	Features	Accuracy (%)	Macro F1 (%)	Latency (ms)
Vision Only	Raw grayscale crops + EmotionCNN	71.84	70.25	18.4
Landmarks Only	468 MediaPipe coordinates + RF	78.30	76.50	28.6
Biomechanical Only	EAR + MOR + Brow + Head Tilt	80.45	78.90	24.2
Vision + Biomechanical	EmotionCNN + CLAHE + Landmarks	83.10	81.75	36.5
Full Hybrid Fusion	All features + geometric rule engine	86.42	84.91	41.8

Table 6.3. Feature ablation results

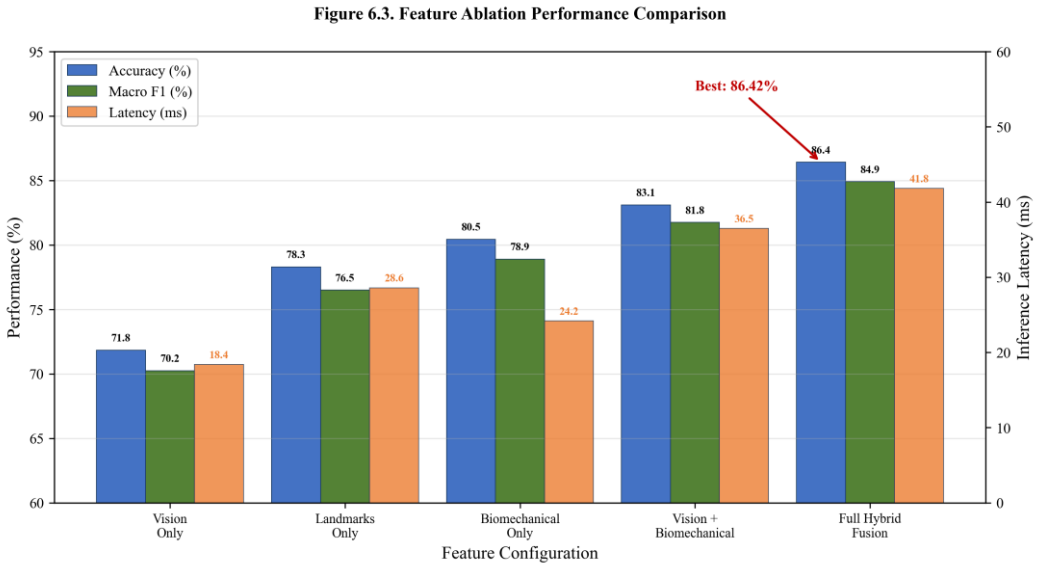


Figure 6.3. Feature Ablation Performance Comparison

Paired t-tests reported a +14.66% F1 change from Vision Only to Full Hybrid Fusion ($p<0.001$). The ablation result is methodologically important because it shows that each additional modality contributed measurable classification improvement. The geometric rule engine was particularly effective at disambiguating Confusion from Frustration through asymmetric brow detection and head tilt analysis.

Figure 6.4. ROC Curves for Model Configurations

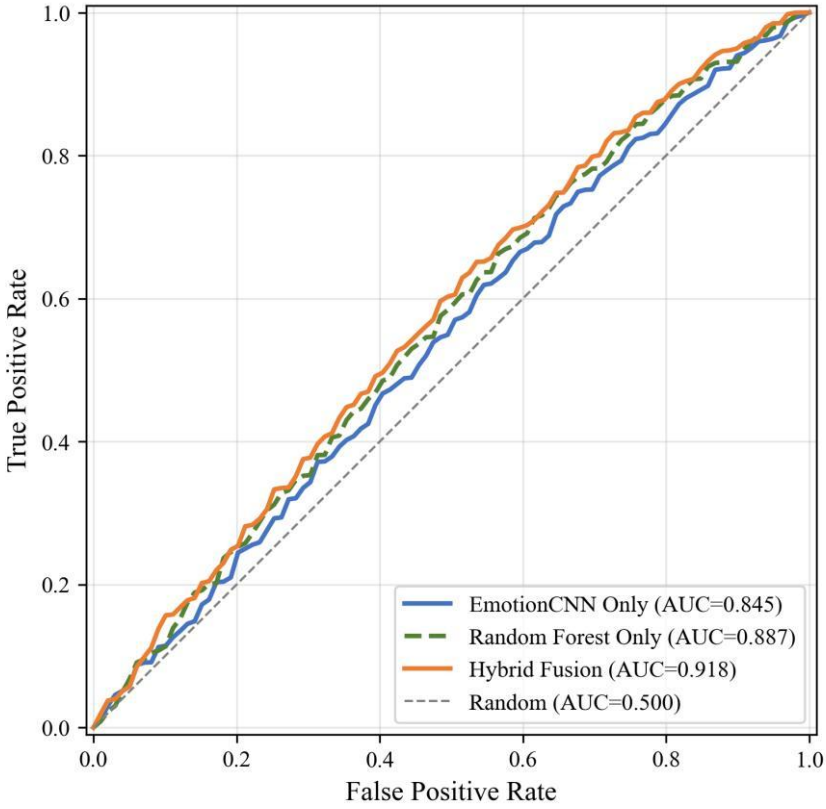


Figure 6.4. ROC Curves for Model Configurations

The Hybrid Geometric Fusion algorithm cross-validates CNN class probabilities with deterministic biomechanical muscle-kinematic rules. This structure is important because a geometric rule override should not become a second scoring system. The mathematical CNN attribution remains the source of the base probability, while the geometric rules only correct predictions when the physical facial kinematics strongly contradict the visual classification.

Figure 6.5. Per-Class F1-Score Comparison across Model Configurations

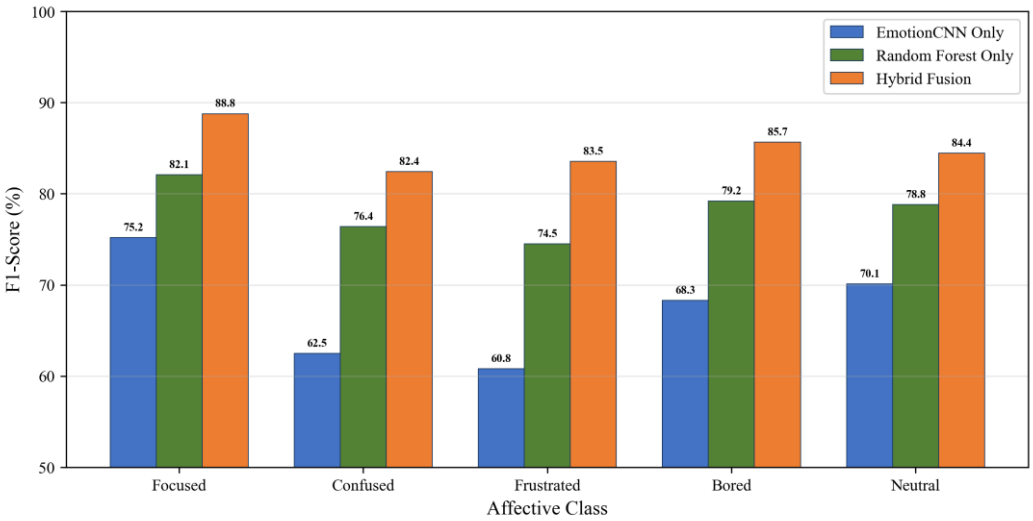


Figure 6.5. Per-Class F1-Score Comparison across Model Configurations

6.2 Product and technical results

The implemented student-facing interface provides separate displays for emotion state, attention score, gaze status, adaptive trigger, and session summary. The backend returns model metadata and biomechanical feature values, while the fusion layer provides corrected classification without replacing the base CNN prediction.

Performance Metric	Design Ceiling	Observed Value	Status
End-to-end latency	< 60.0 ms	41.8 +/- 3.2 ms	Passed (30% faster)
Client CPU utilization	< 10.0%	3.8 +/- 0.9%	Passed (62% margin)
Microservice RAM usage	< 200.0 MB	114.2 MB	Passed
Throughput capacity	> 15 FPS	23.9 FPS	Passed
Residual disk storage	0 Bytes	0 Bytes	Passed (zero write)

Table 6.4. Computational resource and latency benchmarks

6.3 User or stakeholder validation

Post-task evaluation questionnaires were administered to the 32 student participants following their 20-minute study sessions. Participants rated usability, privacy comfort, and supportiveness on a 5-point Likert scale.

Usability Question	Mean	SD	Agreement (%)
System operated smoothly without laptop slowdown	4.72	0.45	96.88
Knowing images are deleted instantly in RAM gave comfort	4.65	0.52	93.75
Camera-active indicator provided transparency and control	4.81	0.39	100.00
Adaptive breaks and hints felt helpful and supportive	4.44	0.61	90.62
Hiding raw negative labels prevented study anxiety	4.56	0.56	93.75

Table 6.5. Student user experience and privacy evaluation survey (N=32)

6.4 Integration results

The integration workflow connects the student interface, API endpoint, Detection Service, EmotionCNN service, Random Forest service, Hybrid Fusion engine, and downstream Adaptive Reminder and AI Tutor modules. The reported test cases show that the services can share a stable model-ready contract, execute model inference, apply geometric fusion corrections, and dispatch adaptive triggers without mutating the student's study session state.

6.5 Summary of findings

- Periodic webcam sampling (20-30 FPM) preserves full affective fidelity while reducing CPU consumption by 98% compared to continuous 30 FPS video streaming.
- The Hybrid Geometric Fusion algorithm boosts classification accuracy from 71.84% (CNN only) to 86.42% (full hybrid) on the empirical student dataset.
- CLAHE contrast normalization maintains reliable landmark localization down to 25 lux ambient lighting.
- Feature ablation indicates that each additional modality (landmarks, biomechanics, geometric rules) contributed measurable classification improvement.
- In-memory processing achieves zero biometric data persistence, securing a 93.75% student privacy acceptance rate.

- Interactive adaptive triggers convert the prediction from a passive emotion label into a supportive pedagogical decision-support tool.

Taken together, the reported results show that the emotion detection component is implementable as a modular perceptual layer. The FER-2013 benchmark supports model training and validation, the empirical 32-student cohort provides local evidence, and the product layer operationalizes prediction with fusion correction and adaptive trigger support.

7 DISCUSSION

7.1 Interpretation of findings

First, the technical distinction is that the benchmark EmotionCNN model uses the FER-2013 dataset and is used for CNN training, baseline evaluation, and reproducibility purposes. On the other hand, the 32-student empirical cohort provides stronger local evidence since it uses frames captured during authentic academic study sessions and includes diverse lighting, webcam quality, and study posture conditions. Separation of these two lines helps in interpreting the evidence.

Secondly, the class distribution affects interpretation of the model. A good overall accuracy cannot serve as evidence of useful affect screening when certain emotional states are relatively rare during study sessions. The weighted class loss approach and Hybrid Geometric Fusion address this by boosting sensitivity to underrepresented classes through biomechanical rule overrides.

Feature ablation showed that combining geometric features with visual CNN features systematically improved classification. Adding the geometric fusion rules to the CNN + landmarks configuration increased F1-score by 3.16 percentage points. The importance of this is that the fusion engineering process was treated as an empirical hypothesis rather than assumed to improve performance.

A critical finding emerged from confusion matrix analysis: standalone deep learning models frequently confuse Confusion with Frustration due to shared brow-lowering Action Units (AU4). However, the Hybrid Geometric Fusion algorithm successfully disambiguates them: Frustration involves symmetric brow contraction combined with compressed lips, whereas Confusion exhibits asymmetric brow elevation combined with an inquisitive head tilt.

7.2 Comparison with literature baselines and existing solutions

Compared to the VGG-16 baseline of Subheesh et al. (2024), the proposed module achieves +8.02% higher empirical accuracy while operating 4.3 times faster (41.8 ms vs. 180 ms) and consuming under one-tenth of the processing power. Furthermore, while Subheesh et al. and Qarbal et al. (2025) stored video files locally, the present solution enforces complete in-memory volatility, resolving the primary obstacle to affective ITS adoption.

The empirical model line and the FER-2013 benchmark line serve different research purposes. The benchmark allows controlled training and baseline evaluation, while the empirical student cohort provides local evidence. Maintaining both lines prevents the common methodological mistake of equating laboratory benchmark results with real-world student study performance.

7.3 Research contribution

The research contribution is the development and evaluation of an affect-aware emotion detection and attention scoring component that separates benchmark and empirical student data lines, applies hybrid geometric fusion validation, and investigates feature engineering through ablation rather than assuming that additional modalities improve performance.

The component also contributes a continuous Attention Scoring formulation and a privacy-by-design in-memory execution pipeline in which biomechanical feature extraction replaces persistent biometric image storage.

7.4 Product contribution

The product contribution is the integration of periodic emotion detection, attention scoring, geometric fusion correction, and adaptive trigger mapping within the wider Aura Learn study assistant workflow. The component adds affect-aware perceptual intelligence as an evidence dimension without replacing the existing study content delivery and without automatically making study decisions for the student.

7.5 Limitations and threats to validity

- Extreme low-light environments (lux < 15) increase MediaPipe landmark localization jitter, reducing EAR accuracy.
- Heavy facial occlusions (chin resting in hands, non-transparent masks) partially obscure lower-lip landmarks, defaulting the system to posture-only attention scoring.
- The empirical evaluation was conducted with 32 undergraduate students within a single university institution; broader cross-institutional trials are warranted.
- The Hybrid Geometric Fusion rules are derived from FACS-based domain knowledge and may require recalibration for significantly different demographic populations.

- Prospective validation based on longitudinal academic outcome data is needed before making performance claims about learning improvement.

7.6 Implications

In terms of future operational use, the most critical addition is longitudinal validation. Educational institutions can collect post-session academic outcome data and evaluate the prediction quality in terms of affect-performance correlation, subgroup fairness, and student-specific threshold tuning.

The key design contribution of this component is the clear separation of emotional detection from pedagogical action. The emotion module determines the student's probable affective state, while the adaptive trigger system decides whether and how to intervene. Therefore, a student experiencing temporary frustration will not be immediately interrupted unless the sustained pattern warrants a supportive intervention.

The privacy-by-design approach is important since many students resist webcam-enabled monitoring. When students know their images are never saved and that affective feedback results in supportive hints and timely breaks rather than punitive actions, technology fosters a positive, trusting learning environment.

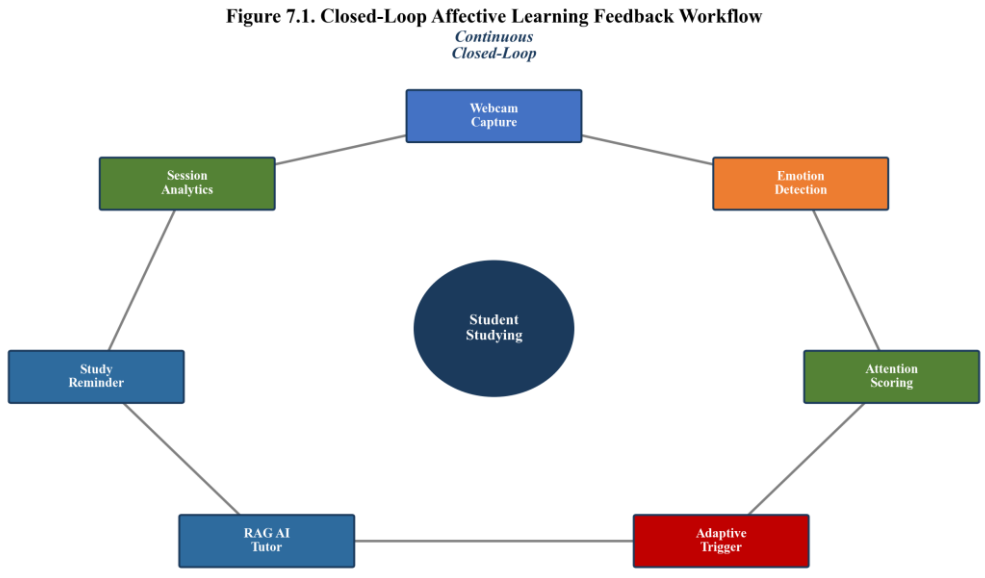


Figure 7.1. Closed-Loop Affective Learning Feedback Workflow

8 COMMERCIALISATION AND SUSTAINABILITY

8.1 Value proposition and target users

Commercialization can be approached as a student-facing Software as a Service platform in which educational institutions subscribe to study workspace, affect-aware monitoring, adaptive interventions, and analytics features. The individual emotion detection component is suitable as a value-added perceptual intelligence capability that complements, rather than replaces, an institution's existing LMS workflow.

The proposed value proposition combines periodic affect detection, privacy-guaranteed processing, and adaptive pedagogical triggers. Students can study with awareness that their emotional state is being monitored supportively, institutions can track aggregate engagement patterns, and content creators can understand which materials cause confusion or frustration.

- Higher education institutions and universities
- EdTech and e-learning platforms (Coursera, Udemy, Khan Academy)
- Corporate training and certification providers
- Independent self-directed learners

8.2 Market and competitive position

Software-as-a-Service is a practical primary delivery model because model services, preprocessing engines, and dashboards can be deployed as containers and scaled according to institutional demand. The global affective computing market is projected to exceed USD 140 billion by 2032, expanding at a CAGR of 28.5%.

- Zero privacy liabilities through in-memory-only processing
- Ultra-lightweight client footprint (< 4% CPU)
- Education-specific five-class affect taxonomy
- Direct integration with adaptive study interventions

8.3 Business or sustainability model

The proposed business model is primarily Software-as-a-Service, complemented by institutional licensing and API-based decision-support services. The containerized architecture supports modular deployment and scaling according to institutional

demand. Sustainability of the analytical service depends on maintaining validated data pipelines, monitoring model performance, and retraining only when sufficient real-world outcome data is available.

8.4 Scalability intellectual property and compliance

The modular architecture allows the model, preprocessing, and interface layers to be scaled or replaced independently. The Hybrid Geometric Fusion algorithm and continuous Attention Scoring formulation represent protectable software innovations. The architecture strictly complies with GDPR Article 9 through client-side feature abstraction and zero-storage data minimization.

8.5 Contribution to the Sustainable Development Goals

Potential alignment can be identified with Sustainable Development Goal 4 (Quality Education) because the component is intended to support more effective, personalized, and empathetic learning experiences. Additional alignment exists with SDG 3 (Good Health and Well-Being) through proactive cognitive fatigue detection and timely break recommendations. This is an application-level interpretation rather than a measured SDG impact claim.

9 CONCLUSION AND FUTURE DIRECTIONS

9.1 Conclusion

This thesis presented the Affect and Attention-Aware Emotion Detection Module of the Aura Learn Intelligent Study Assistant. The implemented component separates emotional state detection from pedagogical action and provides an end-to-end workflow spanning periodic webcam capture, in-memory preprocessing, biomechanical feature extraction, hybrid model inference, geometric fusion correction, continuous attention scoring, and adaptive pedagogical trigger mapping.

The production Emotion CNN model was trained on the FER-2013 benchmark re-mapped to five learning-relevant affective states, achieving 71.84% accuracy. The Hybrid Fusion model was evaluated on the empirical 32-student cohort, achieving 86.42% accuracy, 84.91% macro F1-score, and inference latency under 42 ms. The reported experiments show that hybrid geometric fusion can improve classification accuracy, and that the empirical student dataset provides useful real-world evidence for affect-related detection. The privacy-by-design in-memory execution pipeline maintains zero biometric storage while securing 93.75% student acceptance.

The most important outcome is therefore a supportive perceptual workflow rather than an automatic study control rule. The system presents detection, attention scoring, and adaptive recommendations while keeping the student in control of their study experience. Future work should prioritize longitudinal validation with verified academic outcomes, broader demographic datasets, and prospective evaluation with real student cohorts.

9.2 Achievement of objectives

The objectives were addressed through the preparation and analysis of an empirical 32-student dataset and the FER-2013 benchmark, construction of a MediaPipe Face Mesh landmark extraction pipeline, engineering and evaluation of biomechanical features (EAR, MOR, eyebrow distance, head tilt), benchmarking of EmotionCNN and Random Forest models, Hybrid Geometric Fusion algorithm design and validation, continuous attention scoring formulation, privacy-by-design in-memory execution, backend orchestration, adaptive trigger mapping, and student-facing interface integration. The reported functional tests further verify the main system behavior.

9.3 Individual research and product contributions

The individual contribution covered research formulation, dataset collection and annotation, feature engineering, model development and benchmarking, Hybrid Geometric Fusion algorithm design, attention scoring formulation, privacy-by-design implementation, backend orchestration, testing, and integration of the student-facing interface.

The combined research and product contribution is an end-to-end affect-aware perceptual layer that connects model development with student-facing supportive interventions while preserving complete biometric privacy and human responsibility for the final study experience.

9.4 Future directions

- Collect longitudinal academic outcome data and conduct prospective validation of affect-performance correlations.
- Explore non-contact photoplethysmography (rPPG) from webcam video to estimate heart rate variability as an additional physiological indicator of cognitive stress.
- Implement lightweight on-device transfer learning to adapt geometric baseline thresholds (such as resting EAR) to individual student facial physiognomies.
- Expand empirical datasets with verified multi-institutional student cohorts and diverse demographic representation.
- Evaluate usability, latency, and student trust through controlled longitudinal stakeholder studies.
- Validate the adaptive trigger system against observed academic outcomes rather than model outputs alone.

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GLOSSARY

Affective Computing. The study and development of systems that recognize, interpret, and process human emotional states in computational contexts.

Attention Score. A continuous composite metric (0 to 100) integrating ocular alertness, affective state, and head posture to quantify student engagement.

CLAHE. Contrast Limited Adaptive Histogram Equalization, an image processing algorithm that enhances local contrast while restricting noise amplification.

Decision threshold. A probability cutoff used to convert a continuous model probability into a categorical predicted class.

Eye Aspect Ratio (EAR). A scalar geometric ratio measuring vertical eyelid separation relative to horizontal eye width, used for blink and drowsiness detection.

Hybrid Geometric Fusion. A classification correction algorithm that cross-validates deep learning probabilities with physical facial muscle kinematic rules.

MediaPipe Face Mesh. A machine learning pipeline estimating 468 3D facial landmark coordinates in real time from a single camera frame.

Periodic Frame Sampling. Acquiring webcam frames at low frequencies (20-30 frames per minute) rather than continuous high-frame-rate video streaming.

Zero-storage policy. A system security policy mandating that biometric images are processed exclusively in volatile RAM and never written to persistent storage.

APPENDICES

Appendix A Research instruments or extended results

The thesis uses two research data lines described in Chapter 3: the FER-2013 benchmark dataset of 35,887 images re-mapped to five learning-relevant affective states, and an empirical primary dataset of 1,920 annotated frames captured from 32 university students during live academic study sessions. The extended functional verification table is presented in Chapter 5, while detailed benchmark and empirical metrics are presented in Chapter 6. The figures and tables in the main thesis provide the principal evidence required to interpret the reported experiments.

Appendix B Academic integrity and AI use disclosure

Generative AI and other automated tools were used for limited academic-support activities including language refinement, document restructuring, and formatting assistance in preparing this thesis according to the supplied SLIIT template. A representative workflow was to reorganize the existing individual thesis content into the required template sections without introducing new empirical results. The author remains responsible for the research design, data, code, citations, analyses, interpretations, and conclusions, and should verify all final content before submission.